HackForge: Data-to-Decision Air Quality Platform

Track: Air Quality & Public Health (Pakistan)
Date: October 21, 2026
Target Region: Lahore & Islamabad, Pakistan

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1. Executive Summary & Problem Analysis

Severe seasonal smog and fluctuating PM2.5 levels in Pakistani urban centers present acute health hazards to outdoor workers and morning commuters. Current public information systems suffer from raw data gaps, unverified predictions, and lack of localized contextual guidance. HackForge addresses these challenges through an automated, end-to-end data-to-decision pipeline featuring a three-tier data warehouse, an automated Data Quality Agent, and a zero-hallucination Retrieval-Augmented Generation (RAG) Insight Agent capable of producing cited outputs in both English and natural Urdu.

- **Primary User:** District Health Officers, Municipal Planners, and Citizens.
- **Core Decision:** Issuing timely public health warnings and determining daily safety for outdoor activities.
- **Declaring Baseline:** Persistence model ($t_{tomorrow} = t_{today}$) evaluated against real-time AQI values.

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2. Architecture & Data Orchestration Pipeline

The architecture follows a modular, reproducible pattern using DuckDB and ChromaDB:

1. **Ingestion Layer:** Automated fetching from OpenAQ and meteorological APIs into `data/raw/` with full SHA-256 manifest logging (`manifest.csv`).
2. **Cleaned & Processed Layers:** Unified schema, UTC-to-PKT timestamps, unit alignment, and gold-table generation via `make data`.
3. **Orchestration:** Prefect flows running on hourly schedules managing extraction, quality validation, and ChromaDB vector index updates.

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3. Data Quality Agent

The Data Quality Agent executes non-destructive validation using Pandera and custom heuristics:

- **FIX:** Standardizes date formats, fills missing metadata tags, and corrects minor unit discrepancies.
- **FILL:** Applies linear interpolation to short time-series gaps (≤ 2 consecutive missing hourly observations), clearly tagging values as estimated.
- **FLAG:** Identifies long missing sequences or extreme sensor anomalies without altering underlying values, surfacing warnings directly on the dashboard.

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4. Insight Agent (RAG) & Grounding Mechanics

The Insight Agent ensures strict adherence to facts by decoupling logic generation from empirical data retrieval:

- **Retrieval Store:** ChromaDB persistent vector store containing embedded document chunks and structured metric summaries.
- **Inference Engine:** Groq API using `openai/gpt-oss-20b`.